

Kinds of anomalies : global / t'Hooft anomalies

Where the anomaly lives in this axis?

"small" global anomalies
(t'Hooft anomalies)

Global Symmetry

"large" global anomalies
("global" t'Hooft anomalies)

↑
okay!

(local)
Perturbative detection

gauge anomalies

"small" gauge anomalies
("gauge" anomalies)

Gauge Symmetry

"large" gauge anomalies
"global-gauge" anomalies

Only!
Non-perturbative detection -
(global)

↑
sick!

• Axis 1: What kind of symmetry is anomalous?

1) Gauge Symmetry (redundant) — field A is dynamical.

→ anomaly implies inconsistency of theory

theory cannot be defined as a gauge theory unless anomaly canceled.

2) Global Symmetry (physical symmetry) — field A is a background field

→ anomaly implies you cannot gauge the global symmetry.

• Axis 2: How is the anomaly detected?

1) Perturbatively (local):

→ visible from infinitesimal gauge transf & yields an anomalous divergence / local functional.

(there is also of course a non-perturbative "global" signature for these kinds of anomalies)

- perturbative anomaly is the curvature of the determinant (Pfaffian line bundle) over the space of gauge fields.
 - a non-perturbative anomaly is the holonomy of the same line bundle around loops — topological in nature.
- $\Rightarrow$ if perturbative anomaly is non-zero there is automatically a non-perturbative anomaly.)

1. Gauge + perturbative triangle anomalies

1. 4D chiral gauge anomalies — aka ABJ-type gauge anomaly
come from 1-loop triangle diagrams
 $\rightarrow$ cancel in the SM as Matthew showed

2. $d=2n$ — $(n+1)$ -gon anomalies

$\rightarrow$ fatal unless canceled

- Why people like $SO(5)$.
- Green-Schwarz mechanism — started superstring theory. (Type 1 ST)
gauge, mixed, gravitational anomalies $SO(32)$ or $E_8 \times E_8$ factorize &
a one-loop diagram may be canceled by a tree-diagram
(comes from a one-loop diagram in Superstring theory).
- $PT \sim CP^1$
Twistorial spac. & integrability $F = *F$, Penrose-Ward Correspondence
- All non-renormalizable local counterterms arising from higher

order operators to cancel the anomaly in EFTs.

- Anomaly inflow — anomalous gauge theories in d -dimensions can be made consistent by coupling them to a topological theory in $d+1$ dims.

2. Global + perturbative anomalies ($\frac{n+1}{2}$)-form anomalies aka t'Hooft anomalies.

- Obstructions to gauging global symmetries when a classical symmetry is broken upon being gauged (coupling to background fields) when quantized yet remains valid as a global symmetry.

eg: 1) Flavor symmetry $SU(N_f)_L \times SU(N_f)_R \times U(1)_V$ flavor symmetry of massless QCD has t'Hooft anomaly condense IR below.

2) Chiral Anomaly (ABJ)_{axial} anomaly — global chiral symmetry of a massless Dirac fermion cannot be gauged
 $\pi^0 \rightarrow \gamma\gamma$ decay.

3) Mixed anomalies.

→ Good QFT's but you cannot gauge these.

4) Continuous spacetime symmetry anomalies
— usually global unless you do conformal SUGRA then
Weyl anomaly may be a gauge anomaly, gravitational anomalies

Scale/IR/FI Anomalies (usually good)!

$$1) \text{ YM : } T^\mu_\mu = \left(\frac{D}{4} - 1\right) F^{\mu\nu} F_{\mu\nu} = 0$$

$D=4 \rightarrow$ scale invar classically.

Quantize the theory $\rightarrow$ you get Quantum corrections captured by renormalization.

A regulator will introduce a distance scale

$\rightarrow$ you get RG flow ^{physical} encapsulating changing behavior w/ energy scale $\mu \rightarrow$ scale anomaly! (Good).

$$T^\mu_\mu = \frac{\beta(g)}{2g} F_{\mu\nu} F^{\mu\nu} + \dots$$

* Explains why hadrons exist despite near massless quarks.

* If you were at a fixed pt i.e. $\beta(g)=0$

$\rightarrow$ no scale anomaly $\rightarrow$ Weyl Anomaly in curved spacetime

3. Gauge + non-perturbative (go by the name global gauge anomalies)

- Topological structure of the gauge group allows for non-perturbative violations even if perturbative one-loop anomalies vanish.

- Failure of the partition function to remain inva under

large gauge transformations

ones which are not continuously connected to the identity

eg. 1. $SU(2)$ Witten anomaly where a theory w/ odd # of fermion doublets in 4D is inconsistent b/c $\mathbb{Z}$ changes signs under a large gauge transfer.

analogous in 5 spacetime dims

2. New $SU(2)$ Wang-Wen-Witten Anomaly when a 4D $SU(2)$ gauge theory is coupled to an odd # of Weyl fermions in isospin $-3/2$ or higher rep.

3. Higher dim $SU(N)$ - classified by the homotopy group $\pi_n(X)$ (n^{th}).

eg. 1) 6D $SU(3)$, $\pi_6(SU(3)) = \mathbb{Z}_6$ Witten-like anomaly - cancelled by Green-Schwarz.

2) $SU(N)$ in $4k$ dims can have at most $\mathbb{Z}_2$ anomaly often free in $4k+2$ dim (sd dirac fermions coupled to gravity).

4. Parity Anomaly in $2d+1$ spacetime dims.

* If you have a single massless Dirac fermion in 3D

you cannot preserve both gauge invariance & parity.

→ Must add a Chern-Simons term w/ a half integer coefficient which breaks parity but preserves gauge

↑ invariance → foundational for topological insulators.

4. Global + non-perturbative (Global t'Hooft anomalies).

* Basically anything the previous section but now you want the symmetry to just be a global symmetry.

"Anomalies that approach the identity sufficiently quickly at ∞ " which may not be the connected group $SU(N)$ but 2 disconnected groups.

Parity anomaly — gauge version for QED.

$$\gamma^0 = \sigma_1, \gamma^1 = i\sigma_2, \{\gamma^\mu, \gamma^\nu\} = 2\eta^{\mu\nu}$$

Even dir: $\vec{x} \rightarrow -\vec{x}$

$$\psi(\vec{x}, t) \rightarrow \gamma^0 \psi(-\vec{x}, t), \quad \bar{\psi}(\vec{x}, t) \rightarrow -\bar{\psi}(-\vec{x}, t) \gamma^0.$$

γ^0 transf kinetic term correctly.

$$\gamma^0 (\partial_0 \gamma^0) \gamma^0 = \partial_0 \gamma^0, \quad \gamma^0 (\nabla_i \gamma^i) \gamma^0 = -\nabla_i \gamma^i$$

Since $\{\gamma^0, \gamma^5\} = 0$ ψ_L, ψ_R are exchanged under parity & Dirac mass is P invar.

in odd dir

• No γ_5 , fermion masses still breaks a symmetry: parity

In odd spacetime dir parity is the transf that just flips the sign of one space coordinate since

$\vec{x} \rightarrow -\vec{x}$ is just a rotation

so we define it as $\Psi(\vec{x}, t) \rightarrow \gamma' \Psi(\vec{x}, t)$

$$\bar{\Psi}(\vec{x}, t) \rightarrow -\bar{\Psi}(\vec{x}, t) \gamma'$$

$$w | \quad \tilde{x} = (-x^1, x^2, \dots, x^{2k})$$

$$-\gamma' (\partial_\mu \gamma^\mu) \gamma' = \begin{cases} +\partial_\mu \gamma^\mu & \mu \neq 1 \\ -\partial_1 \gamma^1 & \mu = 1 \end{cases}$$

$\Rightarrow$ Dirac mass term flips sign under parity. No chiral symmetry
to rotate the phase of the mass matrix.

We can have parity invar theory of massive fermions
if they come in pairs of masses $\pm M$, parity will flip
the sign of M while interchanging so the mass term is
invar.

Persists for massless case as well. In $2k+1$ we can write
a Chern-Simons term. For abelian gauge theory this is

$$q \int \epsilon^{\alpha_1 \dots \alpha_{2k+1}} A_{\alpha_1} F_{\alpha_2 \alpha_3} \dots F_{\alpha_{2k} \alpha_{2k+1}}$$

$\rightarrow$ parity violating, invariant upto a total derivative under
gauge transf. st gauge invariance of action is not
spoiled.

← ← mass dim 1.

$$D_\mu = (\partial_\mu - iqA_\mu)$$

F mass dim 2. $\Rightarrow$ CS op mass dim $2k+1 = d$.

$\Rightarrow$ dim less coeff.

Nothing forbids it from appearing so it must be generated radiatively — coefficient $\propto$ explicit parity violating sym.

Use PV regulator since the integral is linearly div

gauge
invar

w/ mass Λ

$$\int \frac{d^{2k+1}p}{(2\pi)^{2k+1}} \frac{\Lambda}{(p^2 + \Lambda^2)^{2k}} \sim \frac{\Lambda}{|\Lambda|} \text{ flips sign when } \Lambda \text{ flips.}$$

so a CS op generated radiatively by PV

$\frac{\Lambda}{|\Lambda|}$ survives when $\Lambda \rightarrow \infty$.

$\Rightarrow$ You get a CS term that breaks parity.